

Transformers

Introduction and Overview

Transformers

Google Scholar

attention is all you need

Articles

About 85,60,000 results (0.04 sec)

Any time

Since 2024

Since 2023

Since 2020

Custom range...

Attention is all you need

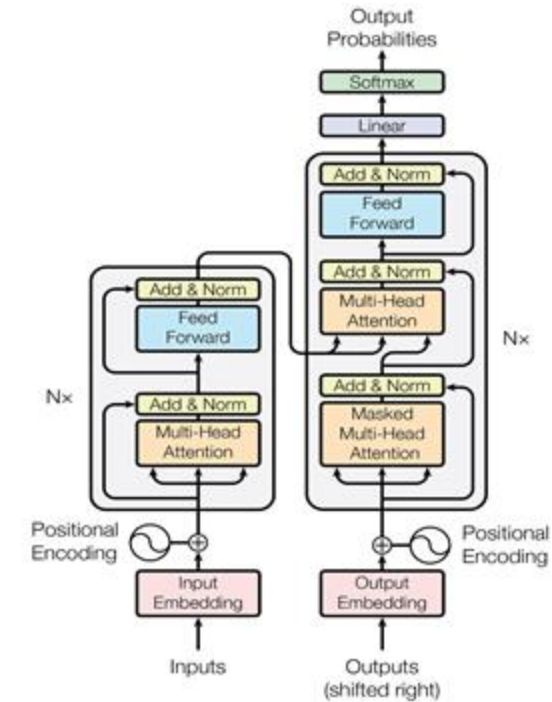
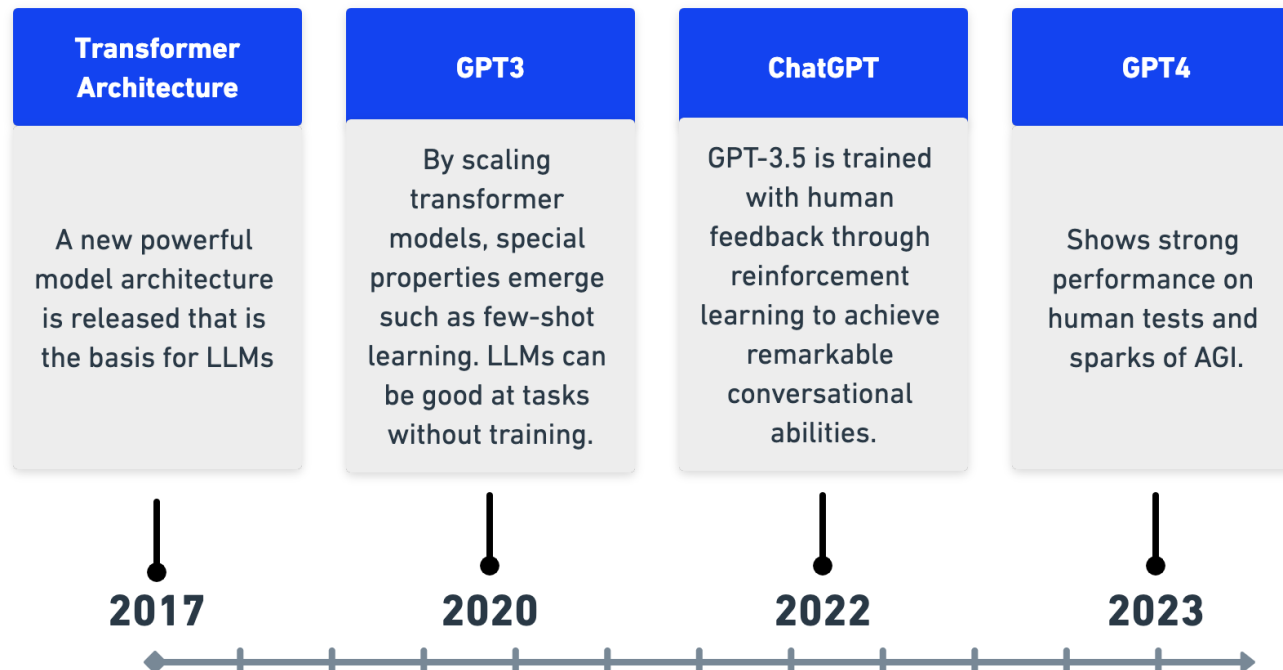
A Vaswani, N Shazeer, N Parmar... - Advances in neural ..., 2017 - proceedings.neurips.cc

... to attend to **all** positions in the decoder up to and including that position. **We need** to prevent ... **We** implement this inside of scaled dot-product **attention** by masking out (setting to $-\infty$) ...

☆ Save 97 Cite Cited by 123714 Related articles All 91 versions 98

[PDF] neurips.cc

Task: Translate from one language to another



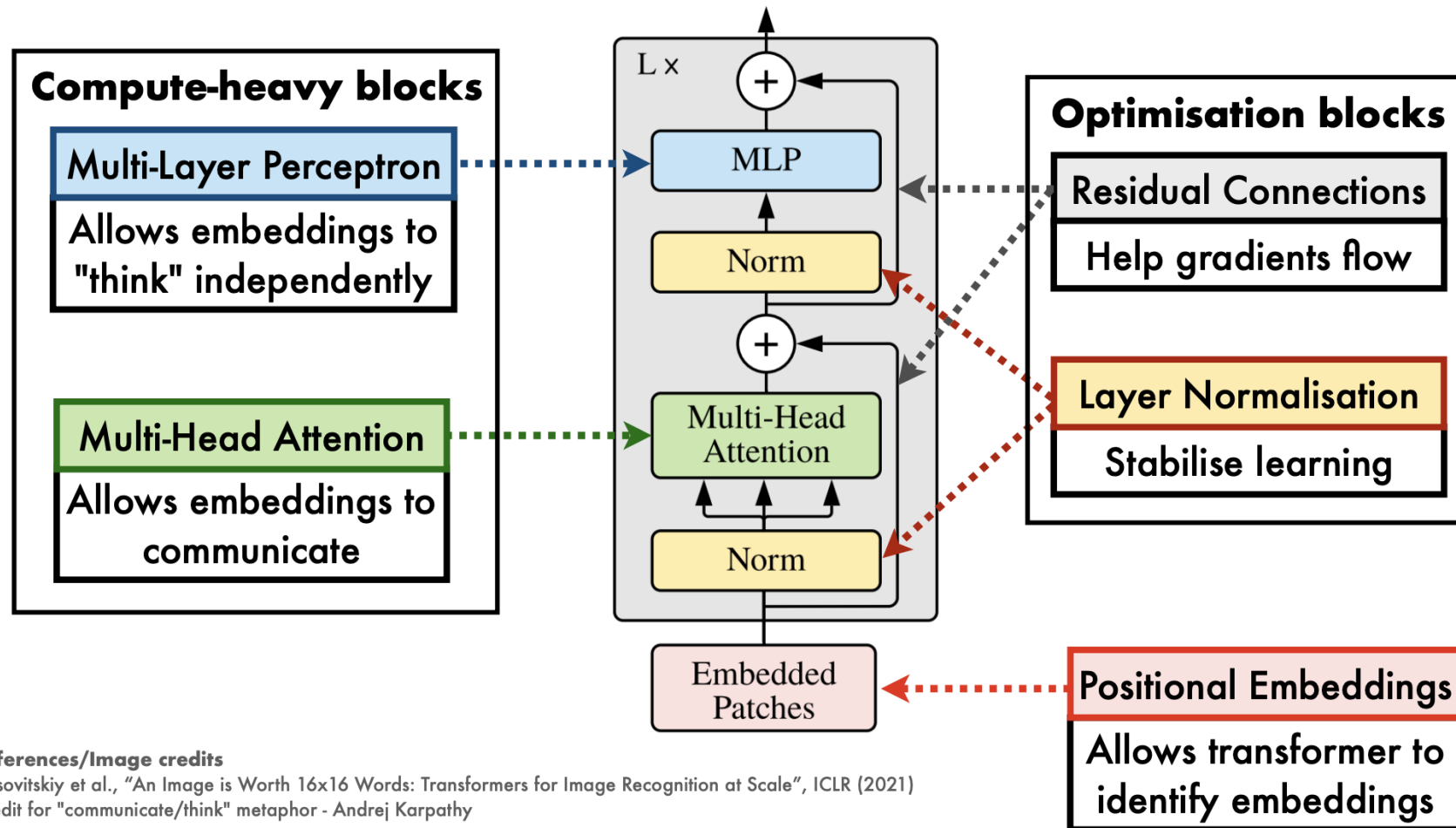
Transformer Encoder
Transformer Decoder

Attention is all you need (2017)

Transformers

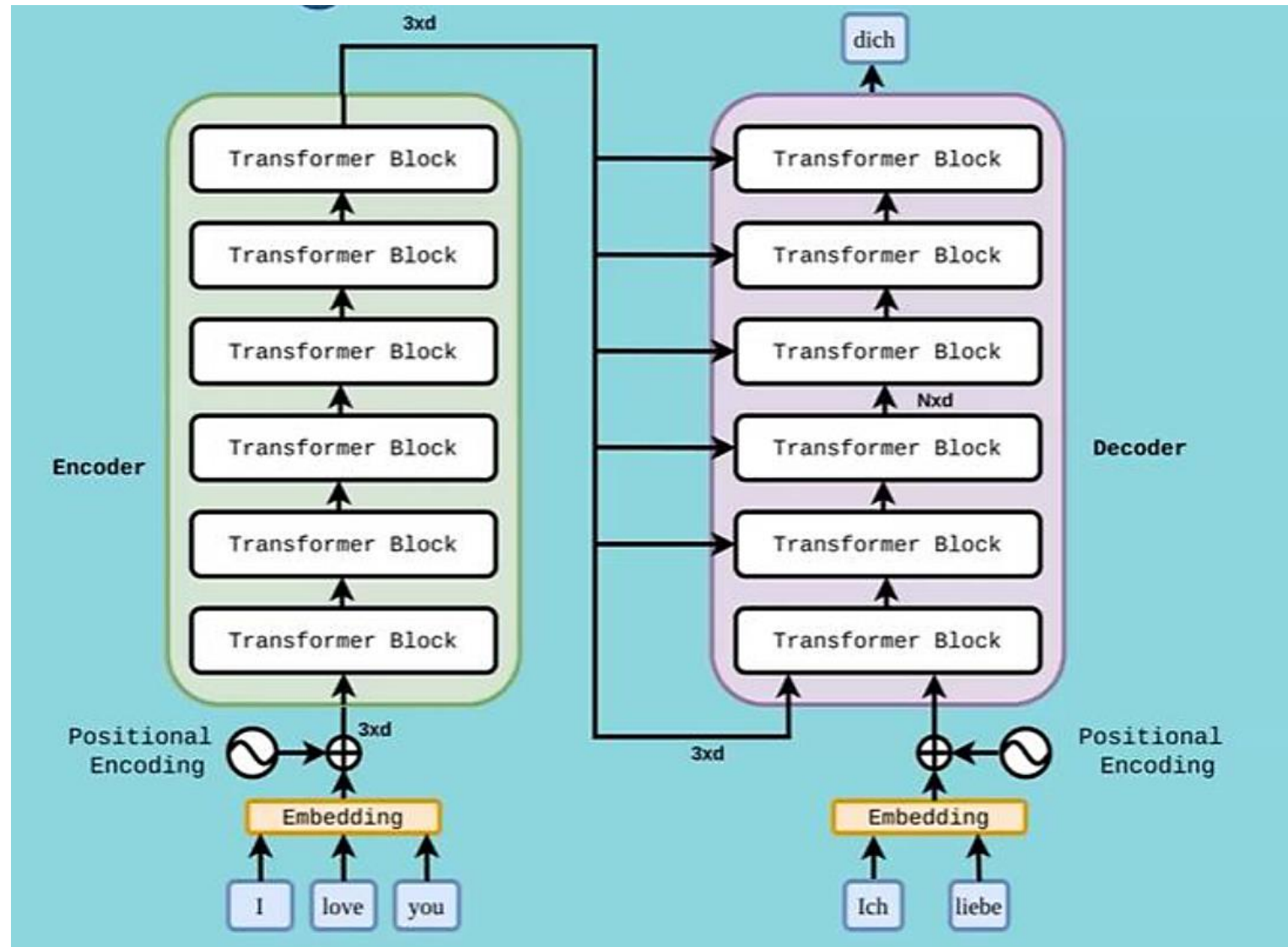
Transformer Encoder

Five key ideas



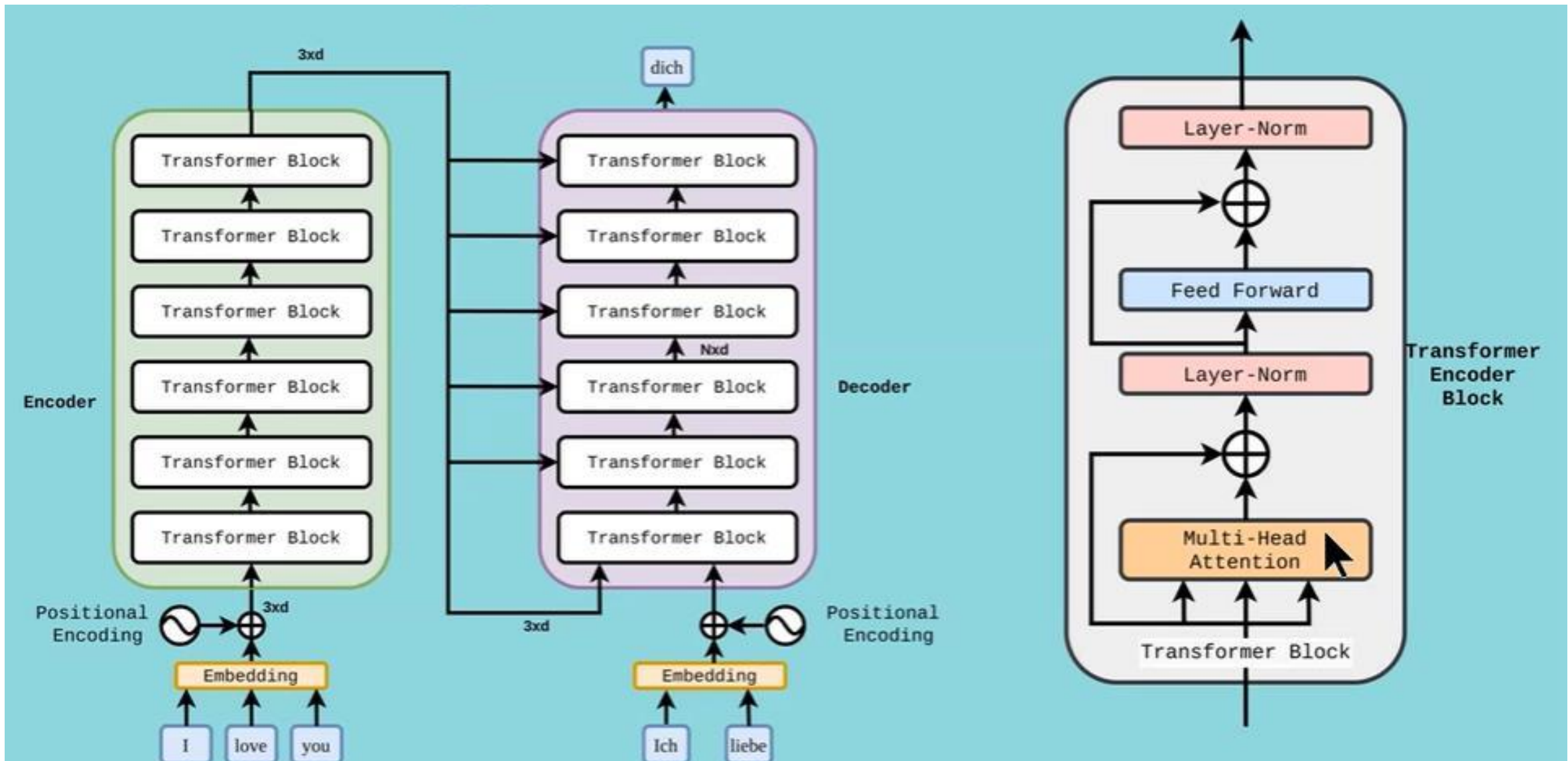
Transformers

Good for
Sets and
Sequences



Encoder only
Decoder only
Encoder-
Decode

Transformer

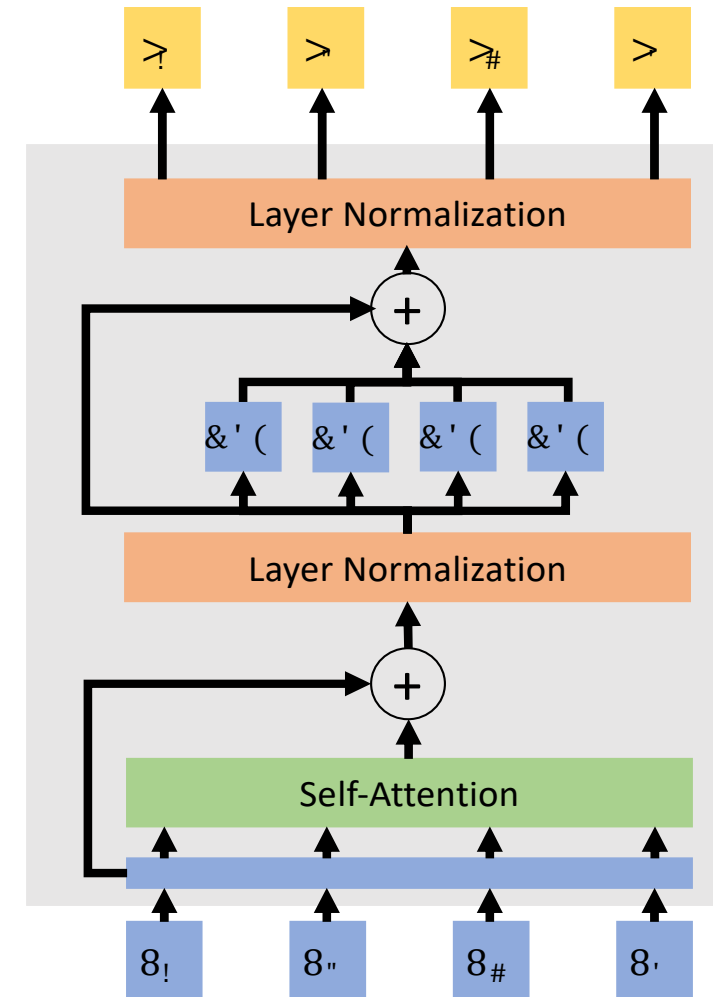


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Building Block

Transformer Block:

- **Input:** Set of vectors x
- **Output:** Set of vectors y
- Self-attention is the only interaction between vectors!
- Layer norm and MLP work independently per vector
- Highly scalable, highly parallelizable



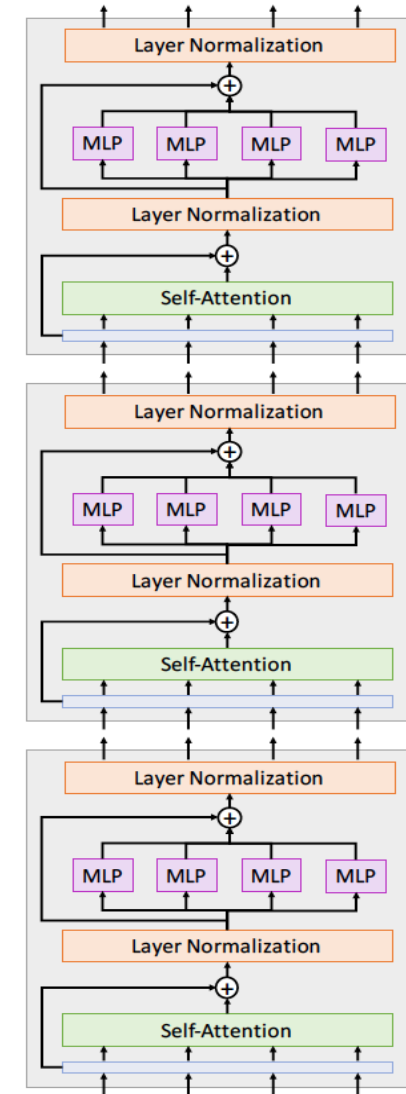
Transformer

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A **Transformer** is a sequence of transformer blocks

Vaswani et al:
12 blocks, $D_v = 512$, 6 heads



Transformer

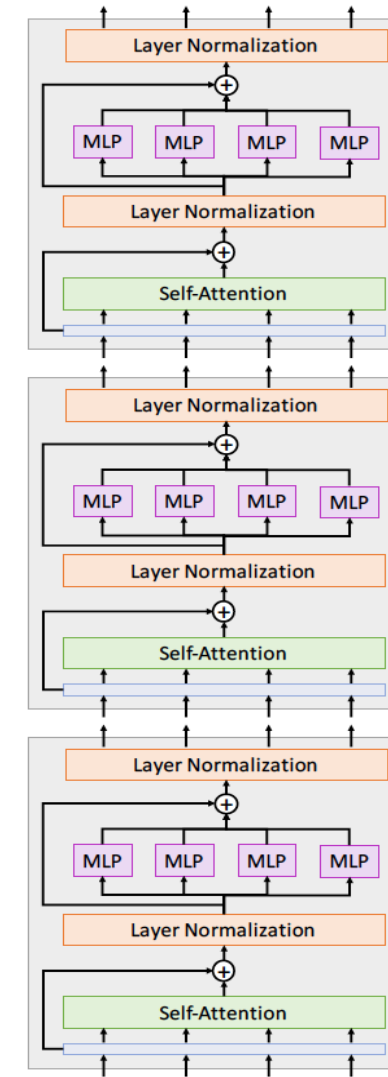
“ImageNet Moment for Natural Language Processing”

Pretraining:

- Download a lot of text from the internet
- Train a giant Transformer model for language modeling

Finetuning:

- Fine-tune the Transformer on your own NLP task



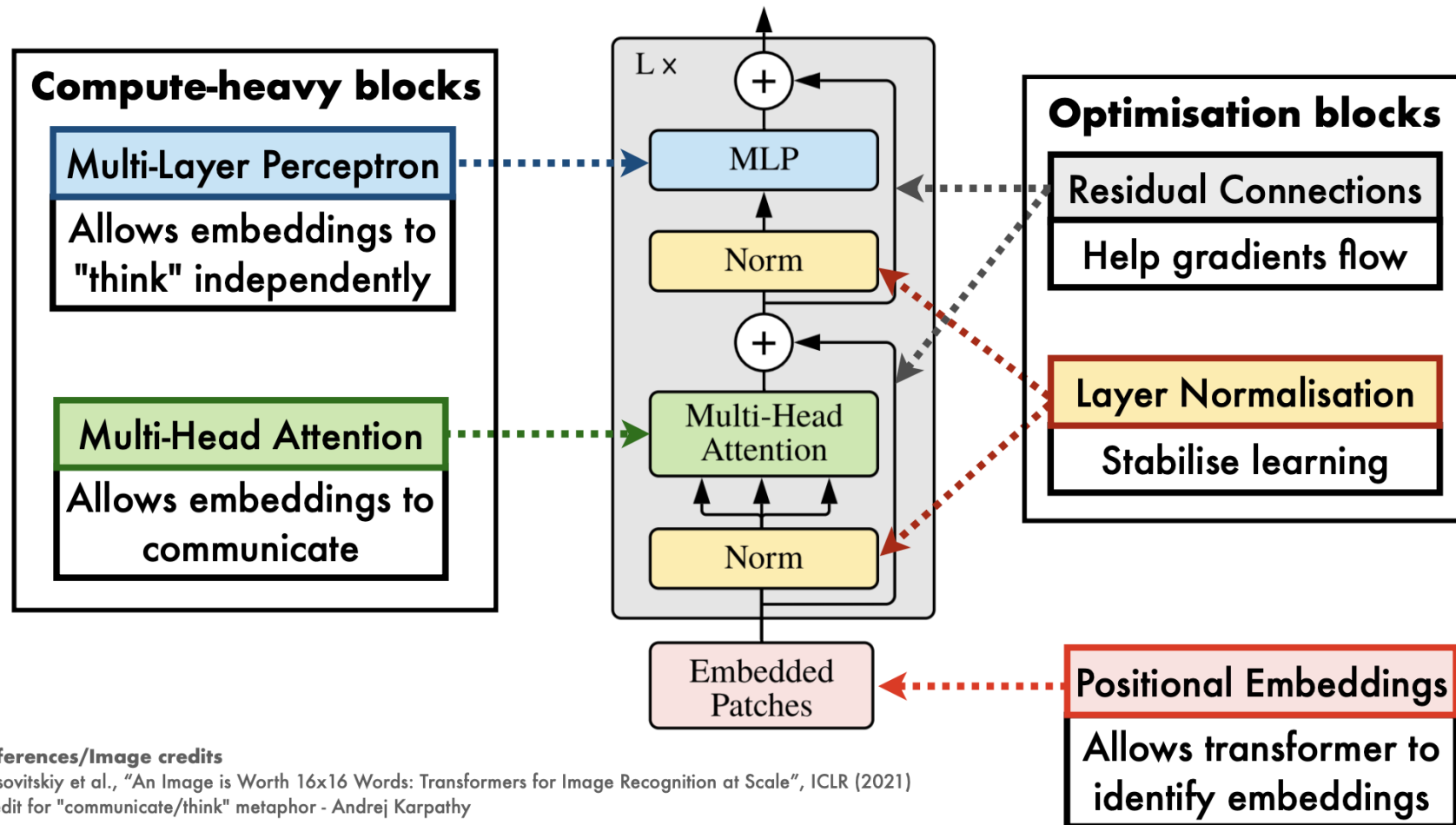
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Transformer

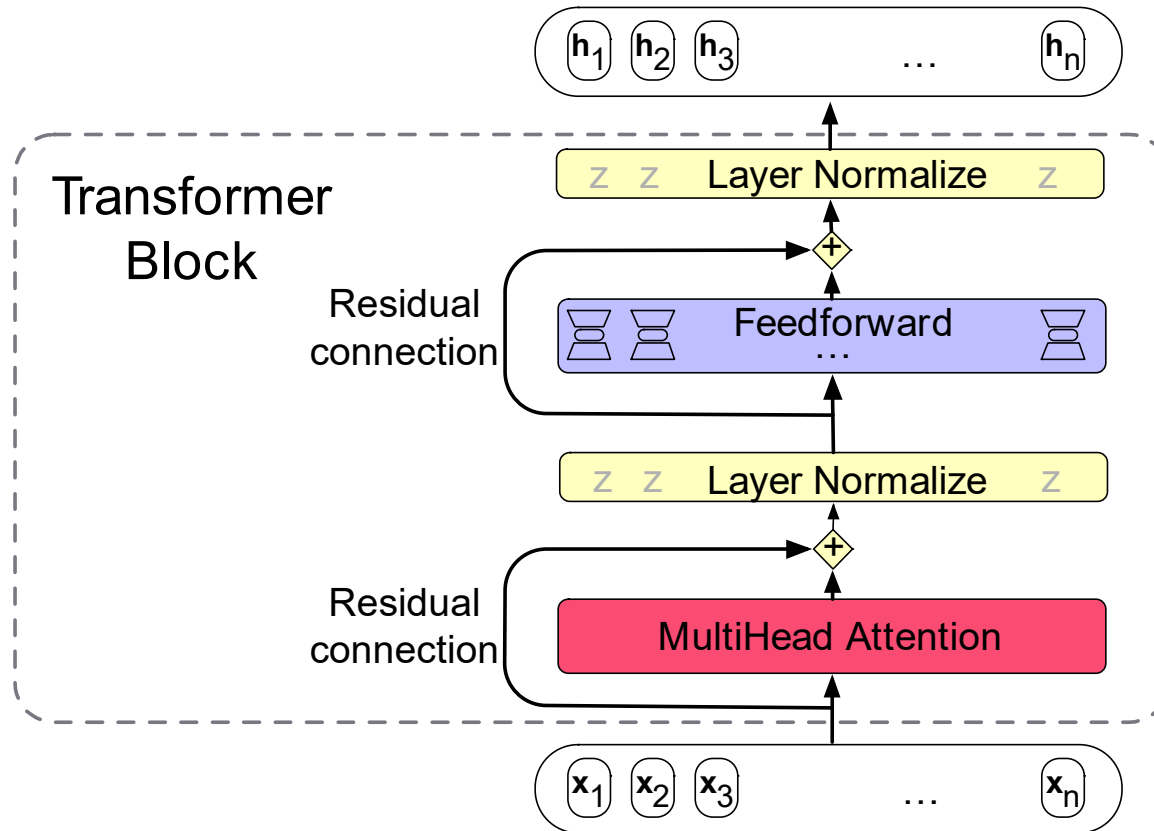
Transformers

Transformer Encoder

Five key ideas



Transformer Block



$$\mathbf{T}^1 = \text{SelfAttention}(\mathbf{X})$$

$$\mathbf{T}^2 = \mathbf{X} + \mathbf{T}^1$$

$$\mathbf{T}_3 = \text{LayerNorm}(\mathbf{T}^2)$$

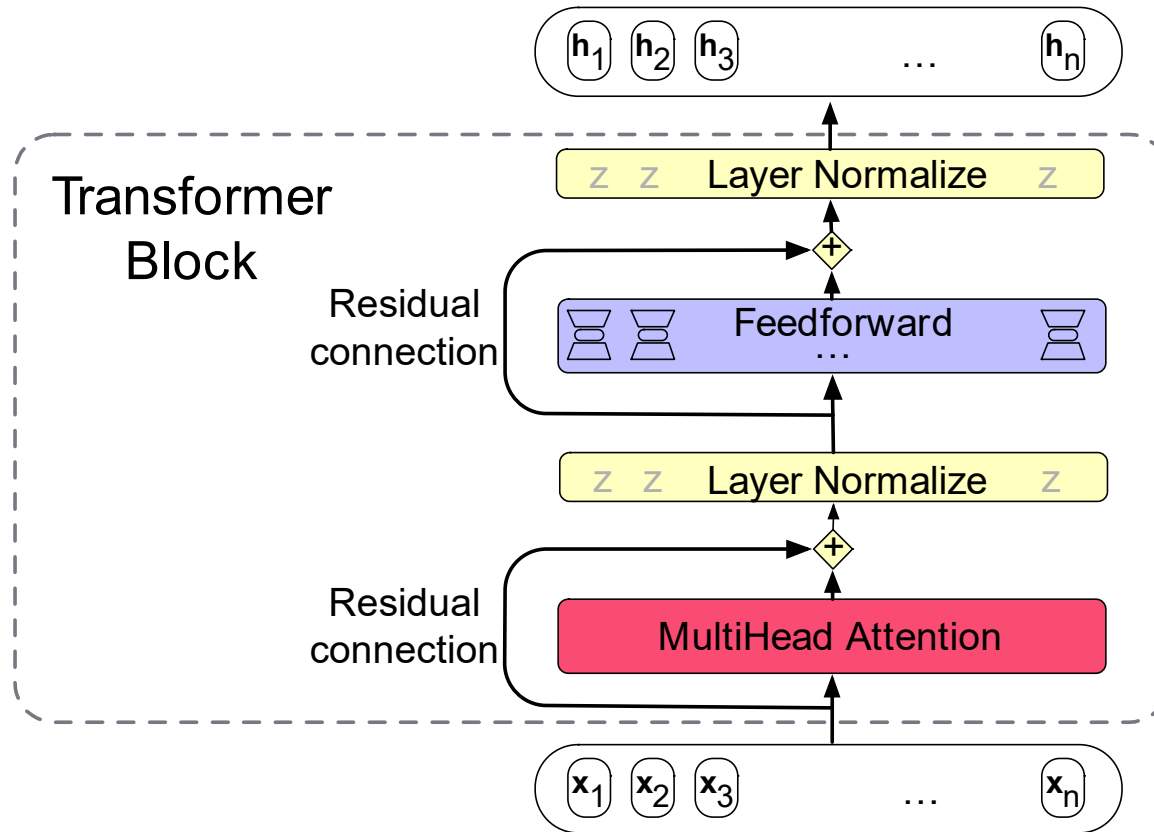
$$\mathbf{T}^4 = \text{FFN}(\mathbf{T}^3)$$

$$\mathbf{T}^5 = \mathbf{T}^4 + \mathbf{T}^3$$

$$\mathbf{H} = \text{LayerNorm}(\mathbf{T}^5)$$

Both X and H are of $N \times d$

Transformer Block



$$\mathbf{T}^1 = \text{SelfAttention}(\mathbf{X})$$

$$\mathbf{T}^2 = \mathbf{X} + \mathbf{T}^1$$

$$\mathbf{T}_3 = \text{LayerNorm}(\mathbf{T}^2)$$

$$\mathbf{T}^4 = \text{FFN}(\mathbf{T}^3)$$

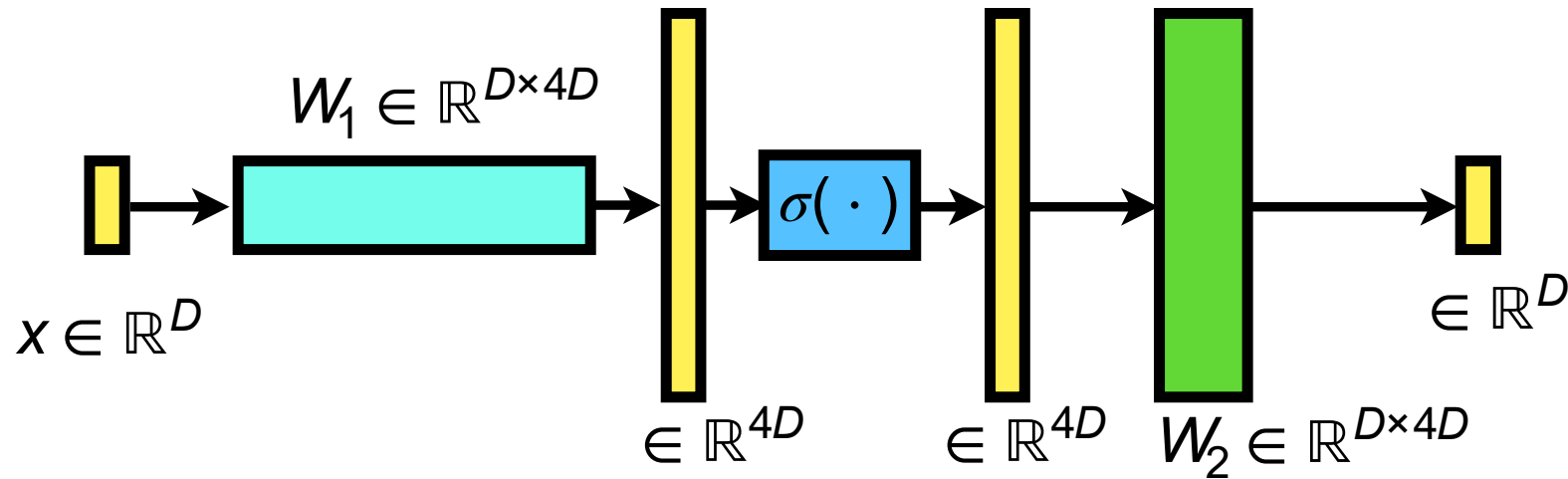
$$\mathbf{T}^5 = \mathbf{T}^4 + \mathbf{T}^3$$

$$\mathbf{H} = \text{LayerNorm}(\mathbf{T}^5)$$

Both X and H are of $N \times d$

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MLP/Feed Forward Layer



$$\text{MLP}(x) = W_2 \sigma(W_1 x + b_1) + b_2$$

Extract combination features from the attended outputs

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Layer Norm Enhances the Stability

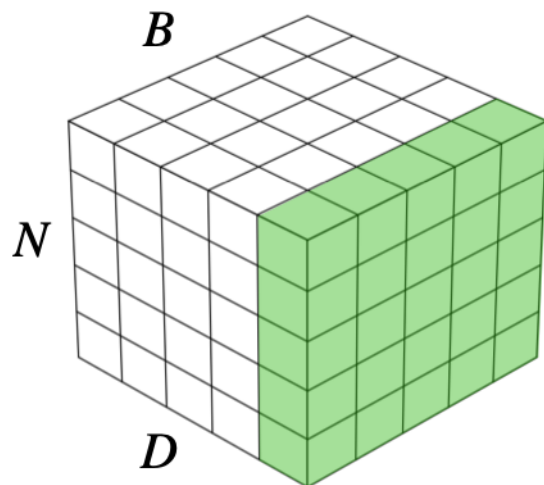
$$\mu = \frac{1}{d_h} \sum_{i=1}^{d_h} x_i$$

$$\sigma = \sqrt{\frac{1}{d_h} \sum_{i=1}^{d_h} (x_i - \mu)^2}$$

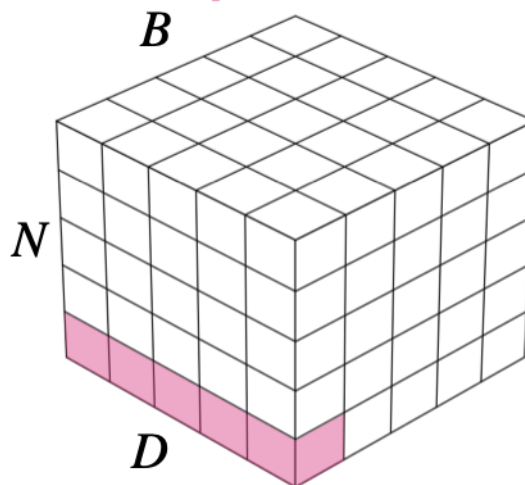
$$\hat{\mathbf{x}} = \frac{(\mathbf{x} - \mu)}{\sigma}$$

$$\text{LayerNorm} = \gamma \hat{\mathbf{x}} + \beta$$

BatchNorm



LayerNorm



LayerNorm has

- No dependence on batch dim.
- Same procedure at train/test time

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Positional Encoding

- Sets vs Sequences=> Strategy: Introduce “Position”
- Position embedding soon after input embedding
- “Bag of words” to “ordered words”

With Positional Encoding

Inputs:

Input vectors: X (Shape: $N_X \times D_Q$)

Key matrix: W_K (Shape: $D_X \times D_Q$)

Value matrix: W_V (Shape: $D_X \times D_V$)

Query matrix: W_Q (Shape: $D_Q \times D_Q$)

Computation:

Query Vectors $Q = XW_Q$

Key vectors: $K = XW_K$ (Shape: $N_X \times D_Q$)

Value Vectors: $V = XW_V$ (Shape: $N_X \times D_V$)

Similarities: $E = \frac{QX^T}{\sqrt{D_Q}}$ (Shape: $N_X \times N_X$) $E_{i,j} = (Q_i \cdot K_j) / \sqrt{D_Q}$

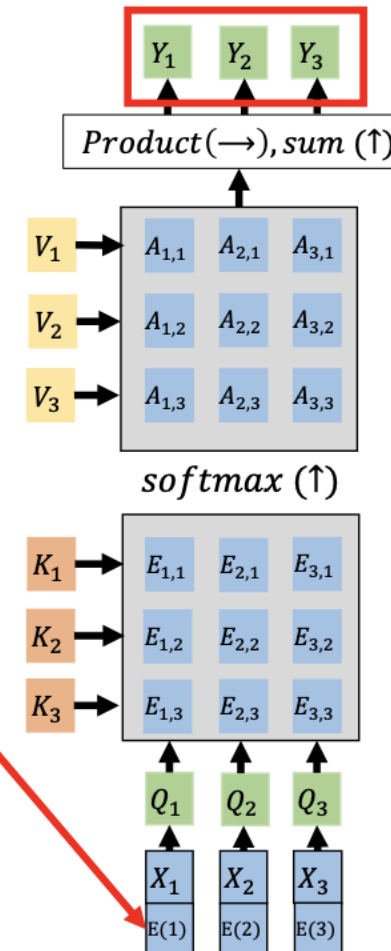
Attention weights: $A = \text{softmax}(E, \text{dim} = 1)$ (Shape: $N_X \times N_X$)

Output vectors: $Y = AV$ (Shape: $N_X \times D_V$) $Y_i = \sum_j A_{i,j} V_j$

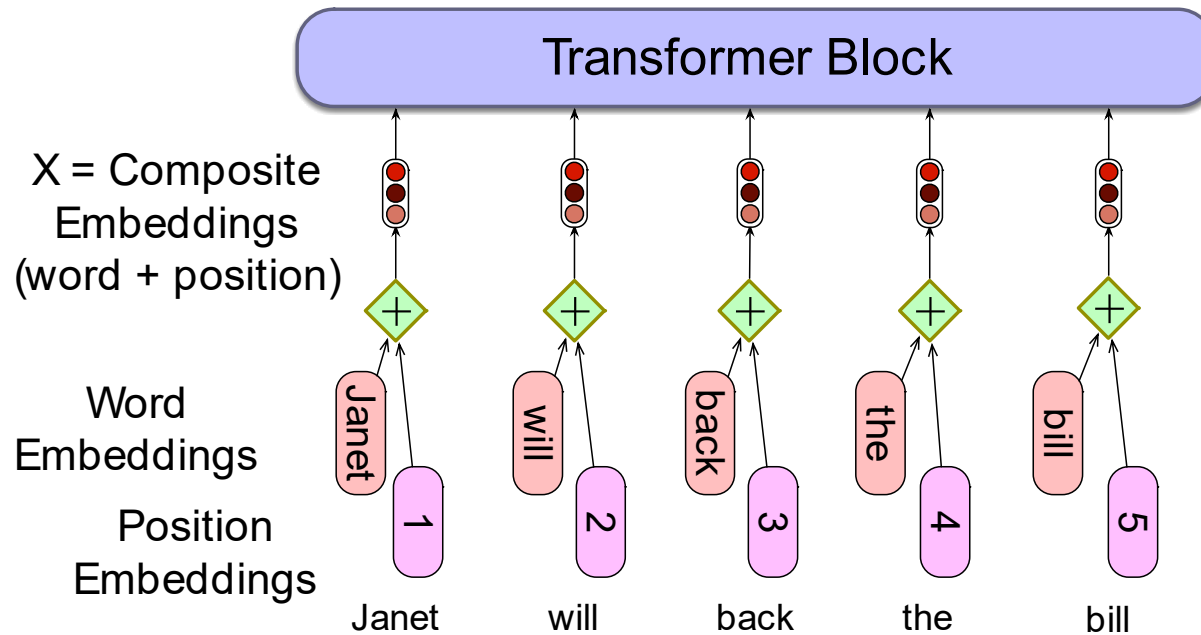
Self attention
doesn't "know" the
order of the vectors
it is processing!

In order to make
processing position
aware, concatenate or
add **positional encoding**
to the input

E can be learned lookup
table, or fixed function



Position Embedding

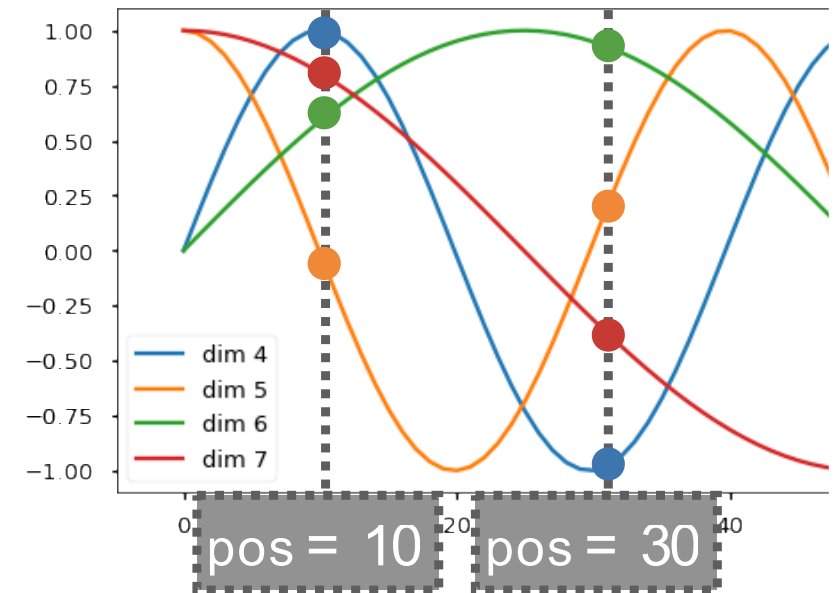


Absolute Positions

Choose static functions (say sinusoids) that map integer inputs to real valued vectors that captures inherent relationship between vectors

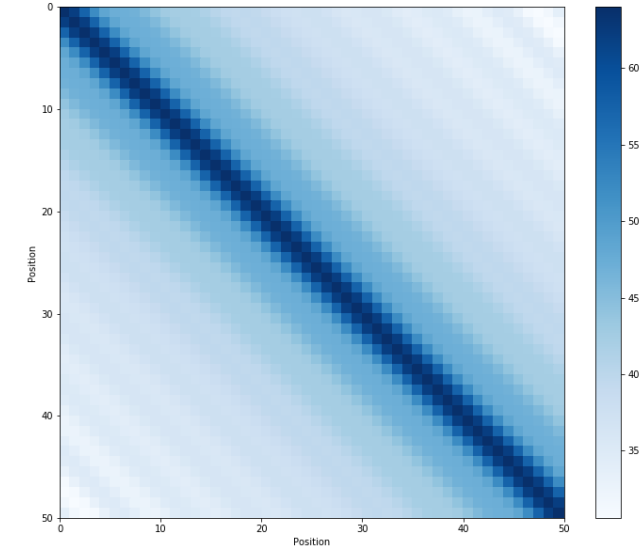
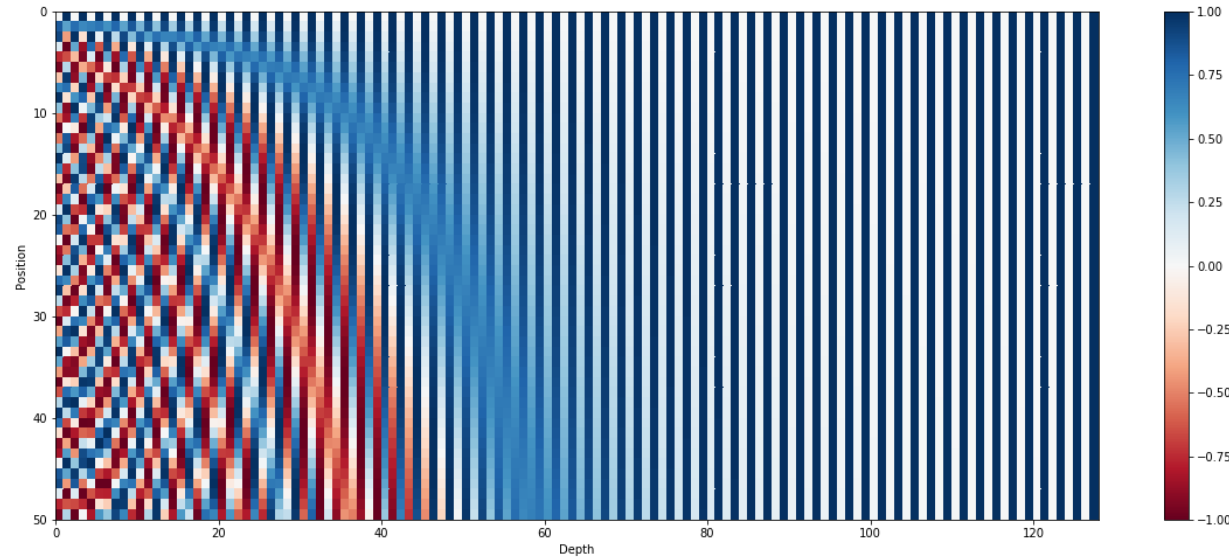
$$PE(pos, 2i) = \sin\left(\frac{pos}{10,000^{2i/D}}\right)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10,000^{2i/D}}\right)$$



Position Embedding

$$p_t^{(i)} = f(t)^{(i)} := \begin{cases} \sin(\omega_k \cdot t), & \text{if } i = 2k \\ \cos(\omega_k \cdot t), & \text{if } i = 2k + 1 \end{cases} \quad \text{where} \quad \omega_k = \frac{1}{10000^{2k/d}}$$



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Many Improvements

- Simplifies LayerNorm by removing the mean and bias terms

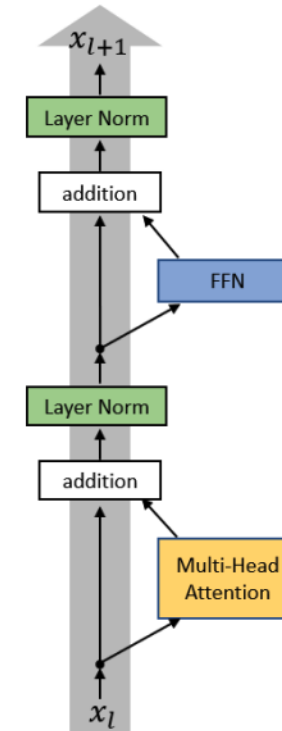
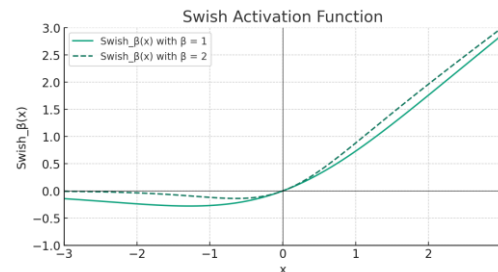
$$\text{RMS}(\mathbf{x}) = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}$$

$$\text{RMSNorm}(\mathbf{x}) = \frac{\mathbf{x}}{\text{RMS}(\mathbf{x})} \cdot \mathbf{g}$$

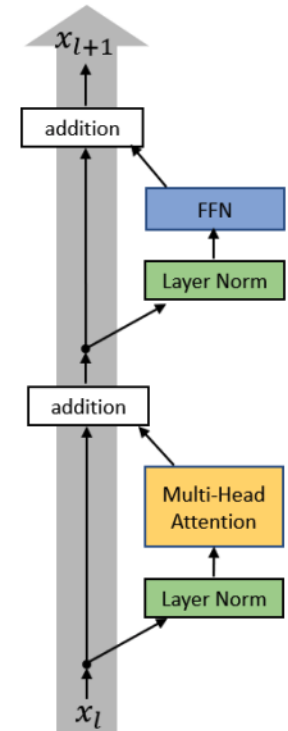
RMS Norm

SiLU

$$\text{Swish}(\mathbf{x}; \beta) = \mathbf{x} \odot \sigma(\beta \mathbf{x})$$



post-LayerNorm



pre-LayerNorm

2017-2024

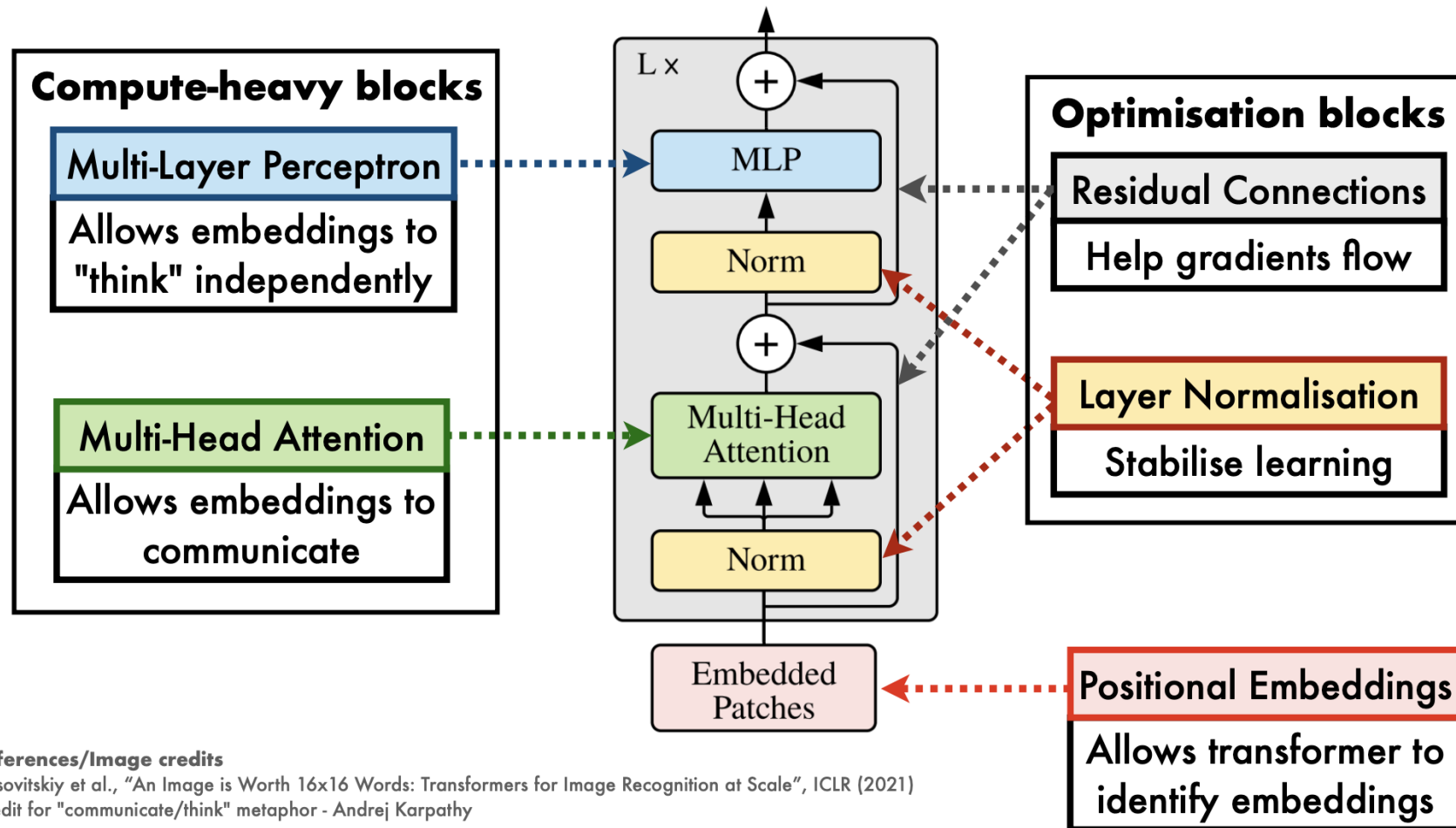
	Vaswani et al.	LLaMA
Norm Position	Post	Pre
Norm Type	LayerNorm	RMSNorm
Non-linearity	ReLU	SiLU
Positional Encoding	Sinusoidal	RoPE

Approximately 10X more efficient (compute vs Performance)

Transformers: Summary

Transformer Encoder

Five key ideas



Transformers: Summary

- Transformers can attend complete sequence and thereby handle long range dependency.
- Transformers
 - Effective use of attention
 - Implementation that can parallelize
 - Scalability to high complexity models and large datasets
- Key Ideas in Transformers
 - Self?Cross Attention
 - Multi Headed Attention
 - Layer Norm
 - Positional Encoding
 - Encoder/Decoder Architecture
- Self Supervised Learning on Large Data

Thanks!!

Questions?